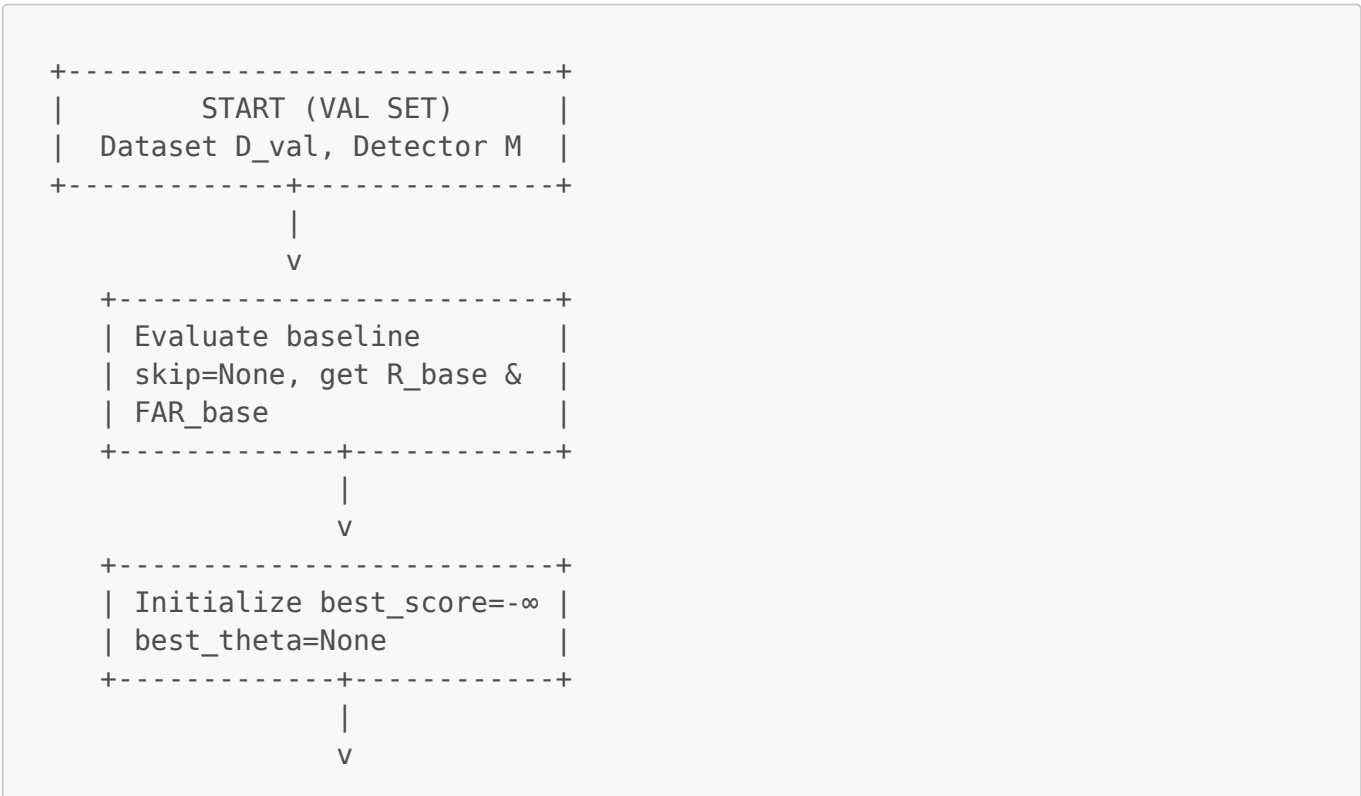
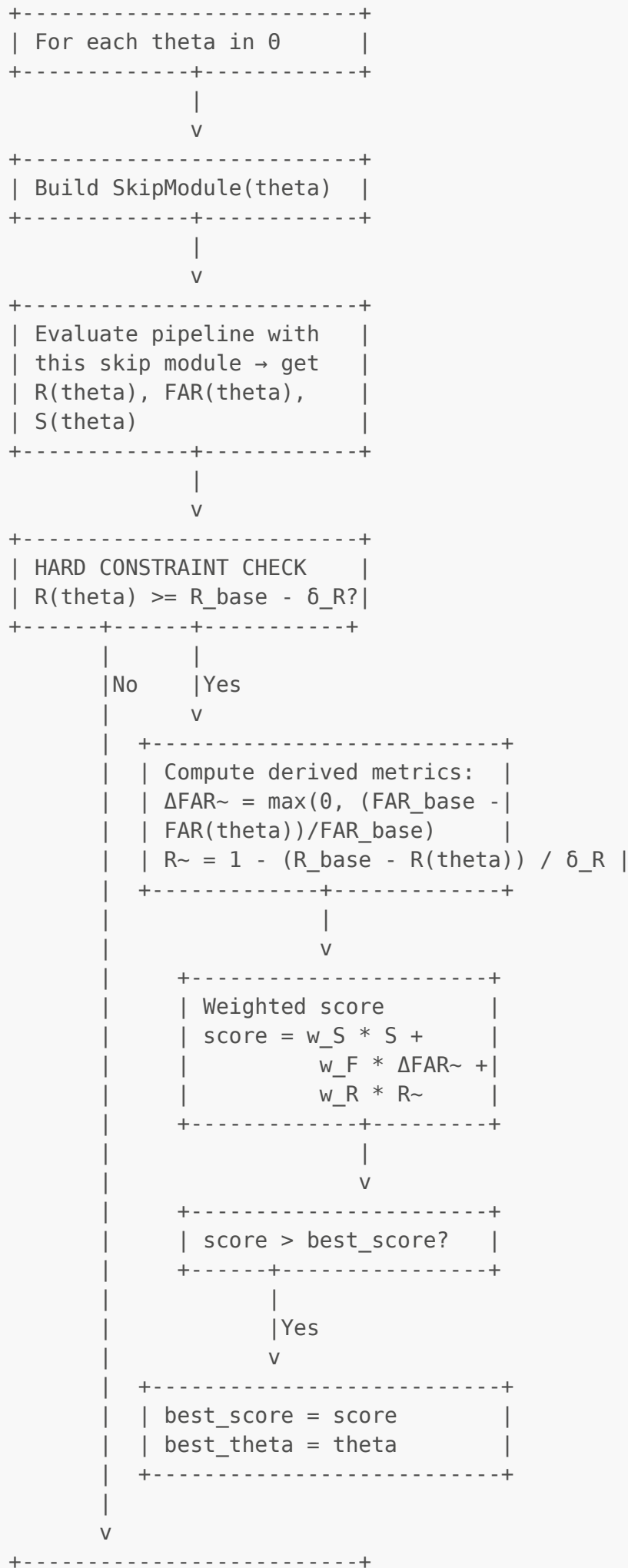


Skip Module Hyperparameter Selection

Symbol	Meaning	Code name
R_{base}	Baseline recall measured with skip disabled	<code>context["baseline"] ["recall"] / r_base</code>
FAR_{base}	Baseline false alarm rate	<code>context["baseline"] ["far"] / far_base</code>
Δ_R	Max tolerable absolute recall drop	<code>context["delta_r"]</code>
$S(\theta)$	Negative-frame skip ratio for candidate θ	<code>df["skip_ratio"]</code>
$R(\theta)$	Measured recall for candidate θ	<code>df["recall"]</code>
$\text{FAR}(\theta)$	False alarm rate for candidate θ	<code>df["far"]</code>
$\widetilde{\Delta \text{FAR}}(\theta)$	Normalized FAR reduction vs. baseline	<code>far_reduction_norm</code>
$\widetilde{R}(\theta)$	Recall-retention term	<code>recall_retention</code>
w_S, w_F, w_R	Objective weights for skip ratio, FAR reduction, recall retention	<code>w_s, w_f, w_r</code>
$\text{score}(\theta)$	Weighted composite score	<code>Combined_Score</code>





```
| NEXT theta in  $\Theta$  |  
+-----+-----+  
|  
  (loop until done)  
|  
  v  
+-----+-----+  
| RETURN best_theta |  
+-----+-----+
```

- Gray arrows indicate the main loop across candidate parameter sets.
- The hard constraint ensures only recall-safe configurations are scored.
- Weighted scoring mirrors Algorithm 1 (docs/param_search.tex).